

Kwang-Hun Lee

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RESEARCH INTEREST STATEMENT

My research interests center on *developing water and nutrient cycle monitoring techniques from local/national to global scales* for sustainable cities and society. Currently, I am keenly interested in developing *uncrewed vehicle-based hydrologic survey techniques* and advancing *explainable AI modeling skill* harnessing remotely sensed data such as Sentinel-2 Mission products.

EDUCATION

Feb. 2022 – 2026 (Expected) Ph.D. in Environmental Engineering (Water Resources Engineering)
Pohang University of Science and Technology (POSTECH), Pohang, South Korea

Advisor: *Prof. Jonghun Kam*

Mar. 2018 – Feb. 2020 M.S. in Earth Sciences and Environmental Engineering
Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea

Mar. 2014 – Feb. 2018 B.S. in Earth Sciences and Environmental Engineering
Gwangju Institute of Science and Technology (GIST), Gwangju, South Korea

HONORS AND AWARDS

2025 Recipient, *Next-generation Convergence Research Pioneering Program*, POSTECH, South Korea
(20 million KRW for one year; University-level student grant competition program)

2024 Outstanding Paper Award, *2023 Korean Society of Environmental Engineering (KSEE) Annual Meeting*

PUBLICATIONS

Peer-Reviewed Journal Articles

5. Lee, K. H., & Kam, J. (2024). Spatiotemporal patterns of water volume and total organic carbon concentration of agricultural reservoirs over South Korea. *Water Research*, 121610. <https://doi.org/10.1016/j.watres.2024.121610>
4. Lee, K.-H., Ali, S., Kim, Y., Lee, K., Kwon, S. Y., & Kam, J. (2023). High resolution mapping of nitrate loads of a reservoir using an uncrewed surface vehicle: Potential opportunities and challenges. *Water Resources Research*, 59, e2023WR034665. <https://doi.org/10.1029/2023WR034665>
3. Choi, S., Lee, K.-H., Park, J.-H., Jeon, S., Lee, H. J., Kang, P.-G., & Kam, J. (2024). Reliable AI models can reveal key processes of heat recovery steam generator operation in air pollutant emission. *Journal of Cleaner Production*, 140806. <https://doi.org/10.1016/j.jclepro.2024.140806>
2. Choi, S., Lee, K.-H., Lee, H. J., Park, J.-H., Kang, P.-G., & Kam, J. (2026). Explainable AI-based optimization of facility operations for excessive NOx emission control: A case study of a solid refuse fuel combustion facility. *Environmental Technology & Innovation*, 42, 104875. <https://doi.org/10.1016/j.eti.2026.104875>
1. Kang, H. Y., Kim, C. S., Kim, D. Y., Lee, Y. J., Park, H. J., Kundu, G. K., Kim, Y. K., Bibi, R., Jang, J. B., Lee, K. H., Kim, H. W., Yun, S. G., Kim, H. Y., & Kang, C. K. (2020). Identifying patterns in the multitrophic community and food-web structure of a low-turbidity temperate estuarine bay. *Scientific Reports*, 10(1), 1–16.

Manuscripts in prep., review, and revision

- **Lee, K. H.**, Nam, W.-H., & Kam, J. Adaptive rice transplanting schedules and reservoir use reveal emerging climate vulnerability in South Korea. *Agricultural Water Management*, *in review*.
- **Lee, K. H.**, Han, S., Oh, H., & Kam, J. High-resolution reservoir monitoring using uncrewed surface vehicles: Spatiotemporal dynamics of nitrate loads and drought impacts. *ACS ES&T Water*, *in review*.
- **Lee, K. H.** & Kam, J., Sentinal-2 *in prep*.

PATENTS

2026 Pollutant Management System and Method (오염물질 관리 시스템 및 방법). Republic of Korea, Patent Application No. 10-2024-0058479.

2023 Pollutant Emission Concentration Prediction System and Method (오염물질 배출 수준 산정 시스템 및 방법). Republic of Korea, Patent Application No. 10-2023-0021417.

SKILLS AND INTERESTS

Python, R, QGIS, Remote Sensing, Machine Learning, Uncrewed Surface Vehicle (USV) Operation, ADCP and YSI EXO2, Uncrewed Aerial Vehicle (UAV) with multispectral band

LANGUAGES

Korean (*Native*) and English (*Advanced*)